

Fueling and Hydrating for the Long Haul

Ultramarathon fueling is tricky. The duration and changing environmental conditions befuddle the nutrition programs of novice and experienced athletes alike. This assertion is supported by the fact that gastrointestinal distress is the leading reason athletes cite as the main cause of DNFs. I've already shown that your ability to physically complete the distance is not a limiting factor for success and that with structured training and smart planning you can overcome the challenges inherent in any ultramarathon course. But all that comes undone when your nutrition and hydration strategies fail. To optimize your performance and have the tools to work through adversity, you need a clear understanding of the fueling challenges specific to ultra-endurance sports, as well as nutrition and hydration strategies that work specifically for you.

Essential Elements of Sports Nutrition

Before we can talk about the nutrition challenges that are specific to ultramarathon athletes, we need to be on the same page in terms of the principles of sports nutrition. There are entire textbooks—and an entire profession—devoted to this topic, but in this book I'm going to cover only the essentials so we can move on to information that's more specific to ultrarunning.

MACRONUTRIENTS AND ENERGY PRODUCTION

Food is made up of three macronutrients: carbohydrate, protein, and fat. All the energy you expend comes from burning calories from these three sources, and you derive energy from all three sources at all times and at all levels of activity. The percentage of energy derived from each macronutrient depends on how quickly you need to produce energy and how much of a supply of each nutrient you have. Sitting and reading this book, you're deriving the vast majority of your energy from fat, although your brain (which is very engaged in learning, right?) is using carbohydrate. Not to be left out, protein is quietly working in the background repairing muscle damage from your last run, helping to maintain your immune system, growing skin and blood cells, and so forth. Protein contributes little to energy production, yet its role is still vital.

When you go for a run or increase your energy demand through any activity, your body responds by ramping up the rate at which macronutrients are broken down into usable energy. During aerobic exercise, energy production happens primarily in mitochondria, which are small organelles in cells (in this case, in muscle cells). Your mitochondria take broken-down carbohydrate and fat and produce carbon dioxide, water, and resynthesized adenosine triphosphate (ATP), the molecule that produces energy. Mitochondria are massively important for endurance performance; having more and bigger mitochondria gives you the ability to process more fuel per minute of exercise.

As you ramp up energy production in response to increased demand, the composition of your fuel mixture changes. The percentage of energy derived from carbohydrate exponentially increases, and this increase continues as exercise intensity rises from an endurance pace to lactate threshold and above. We'll get into the reason for this a bit later. For now, what's important to remember is that while the percentage of energy derived from carbohydrate increases dramatically with demand, the absolute amounts of fat and protein being used for energy also increase, and protein only minimally contributes to energy production.

There has long been a misconception that low-intensity exercise is fueled by fat, high-intensity exercise is fueled by carbohydrate, and there's a magical switch that gets thrown when you cross a certain pace. There's no switch. Absolute fat utilization increases as your exercise intensity increases from a RecoveryRun intensity to an EnduranceRun intensity. As you reach higher intensities, near the upper end of your EnduranceRun or lower end of TempoRun intensities—or 60 to 70 percent of VO2 max, depending on trainability (Maunder et al. 2018)—fat oxidation will begin to decline in order to support exercise at these higher intensities. But there is no all-or-nothing switch from one fuel source to another.

Another reason people misunderstand fat utilization is because they consider it to be a limitless source of available energy, without consideration of the physiological limitations on liberating energy from fat. Although fat is a great source of energy, it takes longer for your body to break down compared to carbohydrate. At rest and at low exercise intensities, you don't use much carbohydrate for energy. In fact, as you sit reading this book, 80 to 90 percent of your energy is coming from fat. At very low exercise intensities (20 to 25 percent of VO2 max), you still rely on fat for about 70 percent of your energy because your body naturally conserves carbohydrate whenever it can. This conservation strategy is necessary because you can store only about 1,600 to 2,000 calories of carbohydrate onboard your body (primarily in your muscles and liver). In contrast, even a lean athlete (70 kg/154 lbs and 10 percent body fat) has more than 53,000 calories of stored fat.

MEASUREMENTS: IMPERIAL VS METRIC

AT TIMES IN THIS BOOK I use imperial units (feet of elevation, miles of distance), and at other times I use metric units (VO2 expressed as milliliters per kilogram per minute or energy expenditure expressed as one calorie per kilogram per kilometer). In this chapter I will be using grams per kilogram to describe the amounts of a nutrient to consume based on your body weight. Why not stick with either imperial or metric throughout? I'm walking the fine line between using common terms for a US audience and correctly representing data used in the sports science and sports nutrition professions. It wouldn't make sense to translate grams per kilogram into ounces per pound or to create an imperial version of VO2 max values.

As you read this chapter, there will be some sample conversions from kilograms to pounds. To determine your weight in kilograms, divide your weight in pounds by 2.2.

As you reach 60 to 70 percent of VO2 max, fuel utilization comes equally from fat and carbohydrate. Better-trained athletes will reach and maintain this fifty-fifty balance at a higher percentage of their VO2 max than athletes who are less fit. As exercise intensity increases above 60 percent of VO2 max, the relative contribution from carbohydrate increases dramatically—but remember, even when you're burning a lot of carbohydrate at higher exercise intensities, you are still processing fat and carbohydrate using aerobic metabolism. For most athletes, once you exceed your lactate threshold intensity (70 to 90 percent of VO2 max, depending on your fitness level), more than 80 percent of your energy is derived from carbohydrate. This, by the way, is one of the fatal flaws of extremely low-carbohydrate nutritional strategies for athletes. When you severely limit the amount of carbohydrate available to working muscles, you are limited by the rate at which your muscles can break down and process fat, which requires more oxygen (approximately 8 percent more than carbohydrate). As a result, you can keep moving forward, but you will be doing so at a slower pace at the same relative intensity.

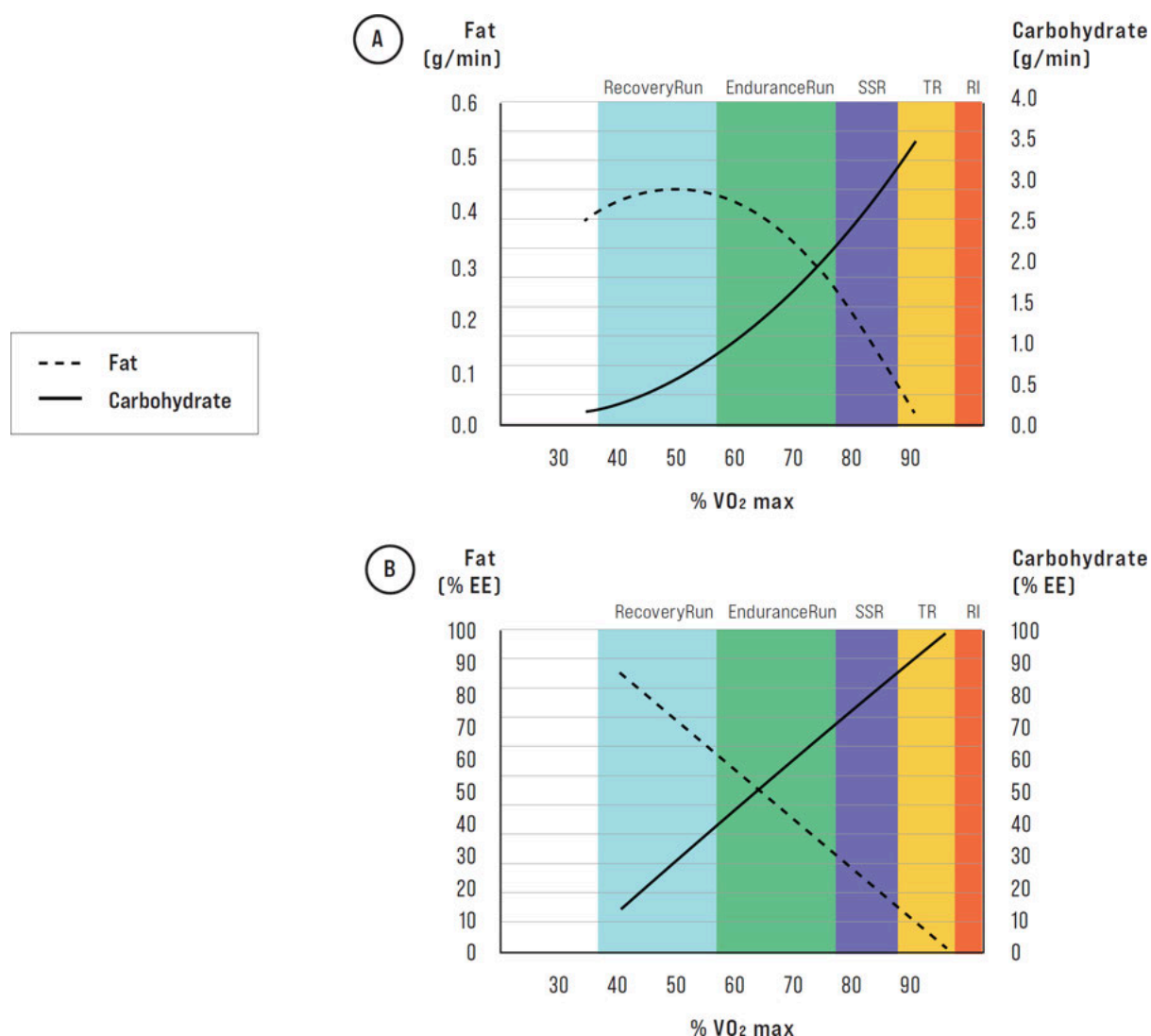


FIGURE 13.1 (A) Fat and carbohydrate oxidation expressed in g/min. (B) Fat and carbohydrate oxidation represented as a percent of total energy expenditure (EE).

As energy demand increases, carbohydrate can be broken down to release a portion of its potential energy quickly. This is called “glycolysis,” and though it yields less energy per molecule of carbohydrate as compared to complete oxidation of carbohydrate, it provides energy to the working muscles faster.

What happens to the leftover metabolic byproducts from glycolysis? Well, the substrate that matters most in this discussion is lactate. This is the stuff that’s unfairly developed a bad rap for years. It has been blamed for everything from the burning sensation in your muscles to delayed-onset muscle soreness. People have tried to massage it away, flush it out, and buffer it. Despite the lack of clarity on the subject, two things have become abundantly clear: (1) lactate is not the cause of fatigue we once thought it was, and (2) the best way to get rid of lactate is to reintegrate it as fuel into normal aerobic metabolism to complete the process of breaking it down into energy, water, and carbon dioxide. Endurance training, particularly when done at a variety of intensities, does this very well. The resulting training adaptations also enable you to recover from hard efforts more quickly, because deriving energy from glycolysis is like buying energy on credit: you have to pay the bill by slowing down, but when you can process lactate faster, you don’t have to slow down as much or for as long.

So, where is protein in all this? Some people push protein as the preferred fuel, perhaps because high school biology taught us that muscle is made of protein, and therefore we need to eat protein in order to use our muscles. Protein is indeed necessary for building and maintaining muscle tissue, but it’s not a very good fuel for exercise. For the most part, it has to be transported to the liver and converted into carbohydrate via gluconeogenesis so it can be transported back to muscles and burned as fuel. Protein plays important roles in sports nutrition (muscle protein synthesis, immune function, production of enzymes, etc.), but those roles don’t include being a primary fuel source. Some protein does get broken down for energy, but regardless of exercise intensity, protein contributes only 10 to 15 percent of your total caloric expenditure.

MICRONUTRIENTS

Vitamins and minerals are micronutrients. Although you don’t derive energy directly from them or need to consume them in quantities anywhere near as large as macronutrients (hence the name), they are essential for your health and performance because of their roles in producing energy, binding oxygen to red blood cells, maintaining bone density, producing muscle contractions, and much more. Because my focus here is sports nutrition for training and competition, in this section I focus on the vitamins and minerals that impact performance most directly.

Micronutrients like calcium, magnesium, zinc, and iron are important. So are vitamins A through K. But during training sessions and competitions, you neither lose very much of these micronutrients through sweat nor use very much to keep going. You are, however, eating quite a bit of food to replenish the calories you are expending. Micronutrient consumption, then, is not much of an issue for ultrarunners, because you are consuming plenty in the foods you are eating, and you are not losing or using much (relatively speaking) in the course of a training session or even an ultramarathon. In terms of performance, a normally healthy person—someone who does not have a condition causing a micronutrient deficiency—won’t really improve performance by focusing on micronutrient intake or supplementation. Your body is already doing everything that needs to be done, provided you have a diverse diet that contains an adequate number of total calories and does not eliminate specific foods or macronutrients.

The Essential Roles of Fluids

When it comes to sports nutrition for ultra-endurance events, fluids are even more important than calories. During exercise, macronutrients have one essential job: to provide energy to working muscles. Water, in contrast, plays a wide variety of roles, and each of them is mission critical.

CORE TEMPERATURE REGULATION

Regulating your core temperature is water's most obvious role during exercise. As you exercise, some of the energy you burn produces the work that moves you forward, but unfortunately even more energy is wasted as heat. This is the price we pay for our overall lack of efficiency. The problem is that the human body operates properly only within a narrow temperature range, from 97°F to 104°F, with temperatures between 97°F and 100°F being optimal for performance (Gleeson 1998). Consequently, a lot of the heat generated from exercise has to be dissipated in order to maintain a core temperature within the optimal range.

Sweat is the body's primary cooling mechanism, with evaporative cooling carrying heat away from the body. As your core temperature rises, sweat glands all over your body start producing more sweat by drawing in fluid from the space around them and secreting sweat onto the surface of the skin. That fluid gets replaced by fluid from your blood plasma, making your blood volume a major reservoir of potential sweat.

As we'll cover in more detail later in this chapter, the amount of fluid an athlete needs to consume depends largely on sweat rate and body size, which can vary greatly depending on exercise intensity, air temperature, wind conditions, and humidity.

GUT MOTILITY AND DIGESTION

Without enough fluid, you cannot break down and absorb food effectively, which means your nutrition strategy is dependent on your hydration status. After food gets broken down in the stomach and travels to the small intestine, the nutrients, fluid, and everything else you want from that food has to be transported through the selective semipermeable membrane that makes up the wall of your intestine. To get carbohydrate from the intestine into the blood, you need to have enough water in the intestine to facilitate the transport. If you don't, the food sits there until enough water becomes available, either because you drink more or because it is pulled from your body into your intestine. This latter mechanism isn't ideal in any circumstance, but it is not a big problem when you are at rest and well hydrated. When you are exercising and pumping sweat onto your skin to cool off, however, your body prioritizes thermoregulation over digestion, and digestion slows dramatically. This is often the tipping point for gastrointestinal distress because once gut motility drops, it can take a long time for it to return to normal, and food that sits in the gut starts to ferment and is jostled around, leading to pressure, bloating, and a cascade of gastrointestinal issues you want to avoid.

BLOOD VOLUME

You have about 4.5 to 5.5 L of blood in your body, and it never stops moving. Athletes and coaches focus on blood's role in delivering oxygen to working muscles, but blood also delivers the nutrients your cells need to function and takes away the waste they produce. Blood also carries heat away from the core to the extremities and skin in order to maintain a healthy body temperature. Finally, blood plasma provides the fluid that ends up being excreted as sweat. One of the key responses to training and acclimation to heat and/or altitude is an increase in blood plasma volume. It's your body's way of filling the reservoir to be prepared for the anticipated activity and environment.

When you run low on fluids and plasma volume drops, your body starts prioritizing how to use what's left. In cold temperatures or in a cold summer rainstorm at high elevation, athletes with a better hydration status stay warmer longer. Dehydration hastens the onset of hypothermia. When it's hot outside and your plasma volume gets a little low, your resting and exercise heart rates increase. Your heart has to pump faster to deliver the same amount of oxygen using less fluid. When plasma volume gets even lower, your body prioritizes sweating over digestion, and if the situation gets dire, it prioritizes oxygen delivery over sweating, and you end up with heatstroke.

WASTE REMOVAL

Removal of metabolic waste products is another crucial role for fluids. This is of particular importance to ultrarunners because of the amount of muscle damage sustained during 50- and 100-mile events. The kidneys filter waste products out of your blood and excrete them in urine. With mild dehydration, urine production diminishes and the color of your urine starts to darken. More severe dehydration can damage your kidneys and alter the pH of your blood.

Day-to-Day Hydration Status

Fluid and electrolyte replenishment during exercise is heavily dependent on sweat rate and temperature. Fluid loss can range from 500 ml/hour to more than 2 L/hour, and electrolyte loss is greatly influenced by the composition of your sweat. Recommendations for fluid and sodium intake become far more complex during training and ultradistance competitions. I will cover those issues in more depth later in this chapter, but to start, let's examine recommendations that will help you stay well hydrated on a day-to-day basis.

Dehydration is often evaluated based on body weight. However, what many athletes don't realize is that they start their day or their workout already dehydrated. A 2 percent loss of body weight during your workout may not be truly 2 percent dehydration, but perhaps 4 percent dehydration if you started the day already low on body fluid.

Researchers Cheuvront and Sawka (2005) devised a simple Venn diagram that is useful for evaluating day-to-day hydration status (Figure 13.2). To use the diagram, you need to evaluate three things immediately after waking up: your weight (W), urine color (U), and thirst (T). If only one observation suggests dehydration but the other two are normal, dehydration is less likely. If two observations indicate dehydration, the condition is more likely. And if all three indicate dehydration, you're very likely to be dehydrated.

First, how thirsty are you? When you are low on body fluid, your body responds with the sensation of thirst. Although there is wide variability from person to person, research suggests dehydration of about 2 percent of body weight is associated with the sensation of thirst (Kenefick et al. 2012). Next comes urine color. The color of your urine is a common ballpark measure of urine concentration. Clear to straw-colored urine is not suggestive of dehydration, but be careful not to automatically equate clear urine with ideal hydration. Urine will often be clear if you are hyperhydrated or hyponatremic, too. If your urine is the color of apple juice or darker, it is suggestive of dehydration. This color observation should also be from your urine stream or a collection cup rather than from diluted urine in a toilet bowl. After urinating, weigh yourself without clothing. From day to day, your weight should remain virtually unchanged. Even if you are gradually gaining or losing weight due to changes in fat or muscle mass, those changes will be very small within a twenty-four-hour period. A loss of 1 to 2 percent of body weight between one day and the next is more indicative of a change in total body fluid than in actual mass.

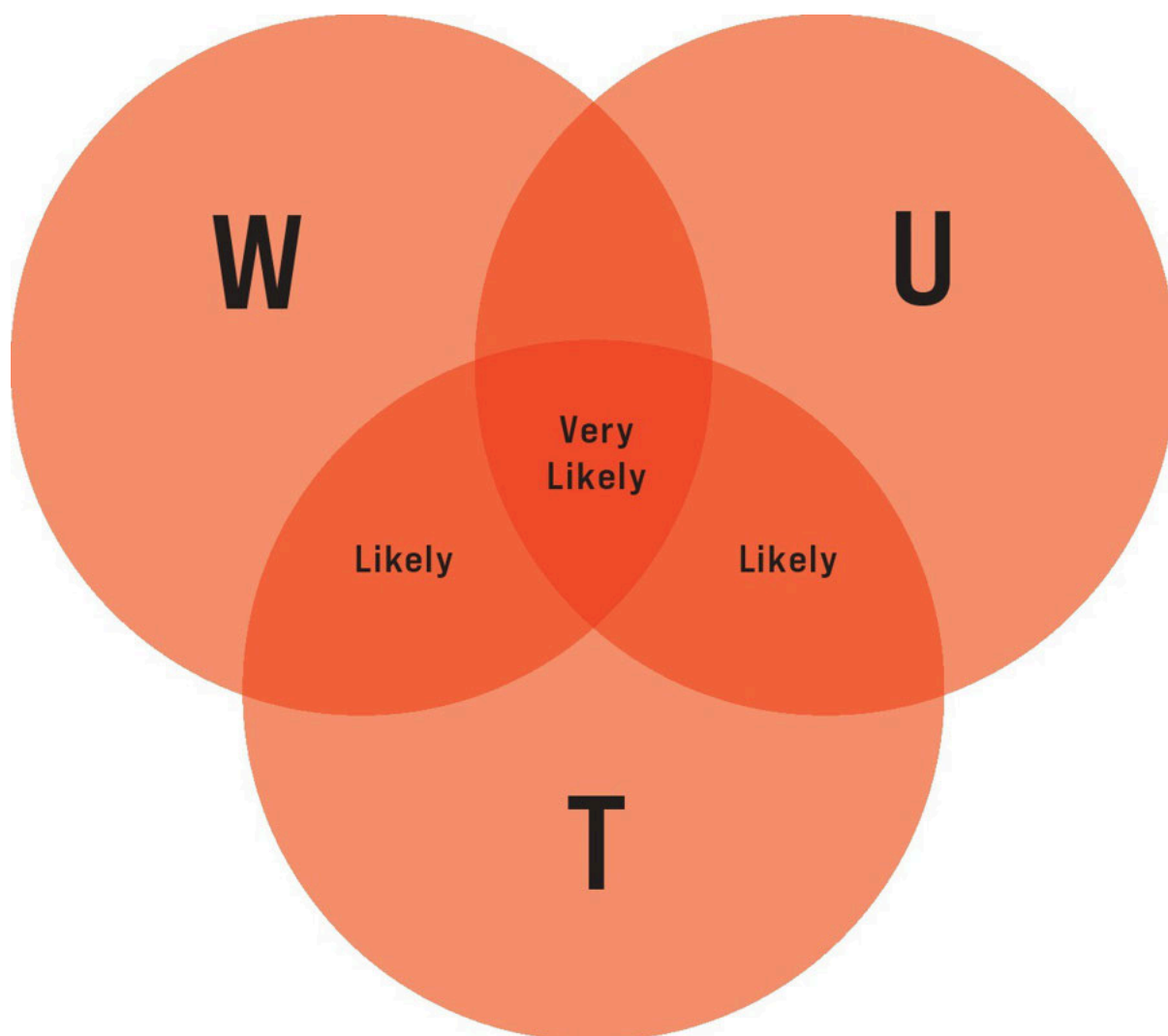


FIGURE 13.2 The WUT diagram helps you monitor your daily hydration status and the likelihood of dehydration. A change in two of the three areas—weight (W), urine color (U), and thirst (T)—indicates that you are likely dehydrated. A change in all three indicates that you are very likely to be dehydrated. Adapted from Cheuvront and Sawka 2005.

If two of the three or all three of these indices for evaluating body fluid indicate dehydration, then you have not done an adequate job of replenishing fluid losses over the preceding twenty-four hours. This doesn't necessarily mean you didn't drink enough during the previous day's workout. It also doesn't mean drinking copious amounts of fluid in the evening is the solution. (This method will increase first-morning urine volume and probably lighten its color, but much of that fluid is likely to pass right through rather than be stored onboard your body and rehydrate you.) It most likely means your overall daily fluid intake was inadequate to replenish combined losses from exercise, thermoregulation during normal activities (sitting outside on a warm day, working in a warm office, etc.), respiration, and water loss due to normal bodily function.

It's typically not a big issue if the WUT diagram indicates minor dehydration one day here and there, nor does one day of minor dehydration necessitate major changes in your habits. It is best to use the WUT diagram over a rolling three-day period. If you are consistently seeing indications of diminished hydration status over two or three days, you should make adjustments to your overall daily hydration habits. This often happens when athletes increase their training load, travel to a warmer climate or higher altitude, or experience a change in weather or seasons. The practical solution is to increase water consumption throughout the day rather than simply guzzling a large volume of water all at once. It is also important to use the recommendations later in this chapter to make sure you're consuming adequate fluids during training sessions.

If you can prevent two of the three (or all three) indices from indicating dehydration, you are starting your day in a better position to maintain ideal body fluid for the rest of the day. You are also more likely to start your workout with adequate body fluid so that when you see a 2 percent weight loss during a run it is an actual 2 percent loss due to dehydration, not a net 4 percent body-weight loss because you started the day and the workout already 2 percent down from day-to-day dehydration.

Sports Nutrition Guidelines for Ultrarunning

The three primary substances you can ingest before, during, and after running to support your activity level are calories, fluid, and electrolytes. More specifically, within these categories, it is most important to ingest carbohydrate, water, and sodium, which, when you distill sports nutrition to its essence, are the three consumables that make

or break athletic performance. The key is to figure out how much you need of each, when you need them, and how to get them.

HOW SPORTS NUTRITION MOVES FROM MOUTH TO BLOODSTREAM

It is easy to get too caught up in trying to determine the ideal formula of a sports drink or the perfect ratio of carbohydrate to electrolyte. What many people overlook is that whatever you ingest gets combined in the stomach, and how it combines determines how quickly it gets into the small intestine, which then determines how quickly it gets into your bloodstream. This is particularly important for ultrarunners because long runs require frequent fueling from varied sources over many hours. The properties of an individual drink or food are less relevant than having a broader understanding of how carbohydrate, fluid, and electrolytes get from your mouth to your bloodstream. So, before we discuss specific recommendations on how many gels to eat, rates of fluid ingestion, and the like, let's take a moment to review how sports nutrition goes from a product wrapped up in a nice little pouch to ultimately fueling your run.

Digestion Starts in the Mouth

Before you even bite into the first piece of an energy bar, your body anticipates the incoming meal and prepares the body to digest it. The simple act of opening up a packet of chews or smelling the cheese quesadillas at an aid station will send signals to your body to anticipate a meal. Your salivary glands will secrete saliva, preparing your mouth for the first chemical process of food digestion, and gastric juices will begin to be secreted into the stomach. As you take a bite, saliva will moisten and to a small extent chemically break down whatever you are ingesting. If required, your teeth further macerate the food into a more swallowable form. All of this is in preparation to pass the foodstuff to the powerhouses of digestion: the stomach and small intestine.

From Stomach to Intestine

One of the primary roles of the stomach is to prepare food and fluid for entry into the small intestine. Your stomach doesn't just break food down from a mashed solid to a liquid slurry; it will hold on to that slurry until the mixture has the right chemical balance before opening the gates to the intestine. The rate at which this happens is called "gastric emptying," and as an athlete you want that process to be rapid.

A few things slow gastric emptying, including highly concentrated foods or solutions (think energy gels or highly concentrated sports drinks). These concentrated carbohydrates sit in your stomach until enough fluid is available to dilute the mixture, which is why it is recommended to consume 8 oz of water with an energy gel.

Taking in too much food and/or fluid can also slow gastric emptying. When you overload your system, gastric emptying can't keep up, and you get a sloshing belly. To a point, increased volume in the stomach accelerates gastric emptying, and you can train your system to go faster. However, even with training there is a limit, which is why consuming smaller volumes of fluid a few times per hour is preferable to guzzling the entire amount needed once an hour.

Absorption from the Intestine

Once the things you've eaten get to the small intestine, it's time to get them into the bloodstream! As with other systems in the body, there are several processes going on at the same time.

The wall of your intestine is a selective, semipermeable membrane, and water and other materials can pass through it in a number of ways. Some things move from the inside of the intestine into the cells of the intestinal wall and then out the other side to the interstitial space between the intestinal wall and the capillaries carrying blood. Other things squeeze between the cells of your intestinal wall into this same interstitial space. When you consume plain water, you are likely to have a low-concentration solution in your intestine, meaning there's a bunch of fluid and not that much stuff dissolved in it. In contrast, the fluid in the interstitial space on the other side of the intestinal wall has a lot of stuff in it. The overall amount of stuff dissolved in a fluid is referred to as its osmolality; in this scenario, the fluid in the interstitial space has a higher osmolality than the fluid in the intestine. Water travels through a semipermeable membrane from an area of low osmolality to an area of high osmolality (it seeks to dilute the high-concentration area). Practically, this means water moves from the intestine to the interstitial space through the spaces between the cells of your intestinal wall, which is what you want.

Whereas water moves passively through the intestinal wall, carbohydrate has to be actively transported. You have distinct channels for different types of sugar: glucose uses door one, fructose goes through door two, and so forth. This is important for endurance athletes because it means you can take in energy faster when you consume multiple types of sugar. Think of it as trying to get into a football stadium: fewer doors mean longer lines and more waiting; more doors mean less waiting. Another benefit to the active transport of sugar is that sugar drags water with it, creating a second method for water to move from the intestine into the interstitial space.

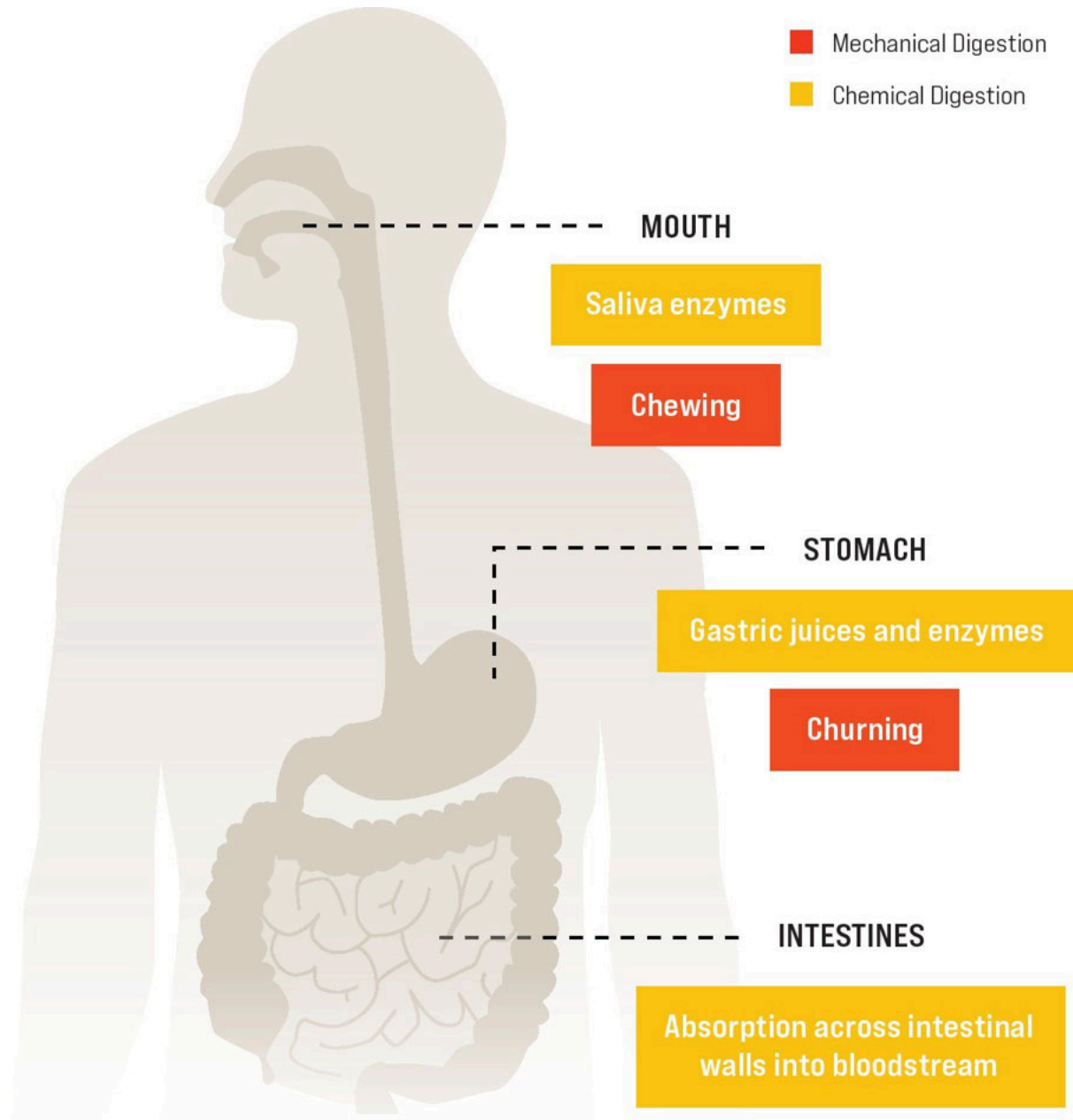


FIGURE 13.3 How and where digestion takes place.

There are also passive and active methods for getting sodium from the intestine to the interstitial space. Like water, sodium can be moved passively through the small spaces between the cells of your intestinal wall, but there are also two active transport methods for sodium. The first is a cotransport with carbohydrate. The cells in your small intestine have gates that enable cotransport of sugar and sodium together (one glucose molecule and two sodium ions) from the intestine into the cell of the intestinal wall. As a result, sodium and carbohydrate move into the bloodstream more quickly when they are present together, which is why even low-calorie electrolyte-rich sports drinks contain some sugar.

Once in the cell, glucose and sodium go their separate ways. Both end up in the interstitial space, but sodium uses a second active transport method: the sodium-potassium pump, which uses energy to pump three molecules of sodium out of the cell and bring two molecules of potassium in from the outside. The sodium-potassium pump is crucial because you are actively moving sodium from an area of low sodium concentration (inside the cell) to an area of higher concentration (the interstitial space), which is the opposite of what would happen normally. This movement of sodium further increases the osmolality of the interstitial fluid, which in turn draws more water from the intestine.

Getting into the Bloodstream

With all these passive and active methods moving fluid, carbohydrate, and sodium from the intestine into the interstitial space, the pressure in that space increases. At this point the pressure gradient becomes the active driver. The water in the space could go two ways: back to the interior of the intestine or forward into the bloodstream. The capillary membrane on the bloodstream side of the space is more permeable than the tighter membrane on the intestinal side. Because water moves from an area of higher pressure to lower pressure, it moves through the capillary membrane into the bloodstream, taking the sodium and carbohydrate with it.

As we discuss the ingredients of sports drinks, recommended foods for race day, and the balance of hydration status and sodium levels later in this chapter, understanding these mouth-to-bloodstream pathways helps you make decisions about the drinks and foods you're consuming.

During-Workout and Race Nutrition

The goal of sports nutrition during a workout or event is to supply your body with the energy, fluid, and electrolytes necessary for optimal performance. Consuming too little of any of them leads to underperformance, and consuming too much will generally lead to gastric distress, which in turn hinders performance. This requires a fine balancing act, one most athletes are inadequately prepared for because they often employ different rates of calorie consumption during events compared to their day-to-day training sessions. For example, they might only consume 100 calories per hour during a long training run yet try to take in 200–300 calories per hour in a race. This is the equivalent of training your legs, heart, and lungs at a 10 min/mi pace on a day-to-day basis and then asking them to operate at an 8- or 6-min/mi pace during the race. That would be absurd, and we need to think about training as not only preparing the legs, heart, and lungs for race day but also the stomach for race-day stressors. In that vein, I am going to present the following nutrition recommendations as strategies for both training and racing conditions.

CARBOHYDRATE

Over the course of the last several years, sports scientists have tried to pinpoint carbohydrate recommendation ranges that can be broadly applied to a variety of endurance athletes. They have studied tolerability of amounts and types of different carbohydrate sources, as well as the body’s ability to use the ingested foodstuffs. The sum total of it all can be distilled down to a recommendation to consume 60 to 90 g/hour of carbohydrate for endurance activities lasting more than ninety minutes. This recommendation is based on the fact that you are able to absorb and utilize about 1 g of exogenous carbohydrate per minute during exercise. With training and a combination of carbohydrate sources (glucose and fructose), you may be able to increase this to about 1.4 g/min, or about 84 g/hr.

There is some emerging evidence that, with specific gut training, rates of over 90 g/hr can be utilized and could be of additional benefit (Costa et al. 2017; Cox et al. 2010; Urdampilleta et al. 2020). However, the utility of such high rates of carbohydrate intake is likely limited in an ultrarunning situation because the energy demand side of the equation does not require it. At ultramarathon distances the rate of caloric output, as well as the percentage of carbohydrate being utilized, are both less than what they would be in the more traditional endurance circumstances upon which many of these guidelines are based.

To get a better fix on this, let’s do a little simple math based on some broad generalizations about ultramarathon intensity and the percentage of fat vs carbohydrate you would utilize in an ultra. In just about any ultramarathon, you are going to be running at an intensity of <65 percent of your VO2 max. Sure, elite athletes can do the climbing portions of a 50-miler at slightly higher intensities, and maybe if you were trying to set a 50K world record you would be at 70 percent of your VO2 max, but stay with me here. Earlier in this chapter I described the “crossover concept,” which is a widely used physiological concept that describes the point at which you are deriving 50 percent of your calories from fat and 50 percent from carbohydrate (see Figure 13.1). A crossover point of 65 percent of VO2 max is quite common for trained endurance athletes. (Plus, it makes the math easy in an ultramarathon, and I like easy math.) This means that at a normal ultramarathon intensity, roughly 50 percent of your fuel is coming from fat, of which you have a nearly unlimited reserve, and 50 percent is coming from carbohydrates, which you will need to replace. You could extend these percentages by ~10 points on either end based on the situation, but the gist would remain the same.

Now, how many calories are you burning per hour? This is actually a much bigger variable. It’s based off of body weight and speed, but if we make some simple assumptions based on a normal person running at a normal intensity, we can once again get a general fix on the situation. A 70 kg person (154 lb) moving at 9:40 min/mi (10 K/hr) on flat terrain is going to utilize ~700 cal/hr. Since only 50 percent of these calories are derived from carbohydrate, this normal ultramarathon pace for a normal runner is going to cost them 350 cal/hr of carbohydrate based on our earlier assumptions. Now take into consideration that you have ~2,000–2,800 calories of glycogen and blood glucose on board, provided that you are in a fed state (i.e., not fasting or doing a run immediately after waking up). This means that this normal runner running at a normal ultramarathon pace can run for over five hours and forty-two minutes or fifty-seven kilometers without any fuel. Increase the runner’s size or pace, and the duration/distance goes down. Decrease the pace or size, and this goes up. Am I telling you to go out and run fifty-seven kilometers without food? Hell no. Please don’t. That’s stupid. But the simple math demonstrates that because most ultramarathon intensities are so low (<65 percent VO2 max), higher rates of carbohydrate ingestion (up to 90 g/hr) are likely unnecessary.

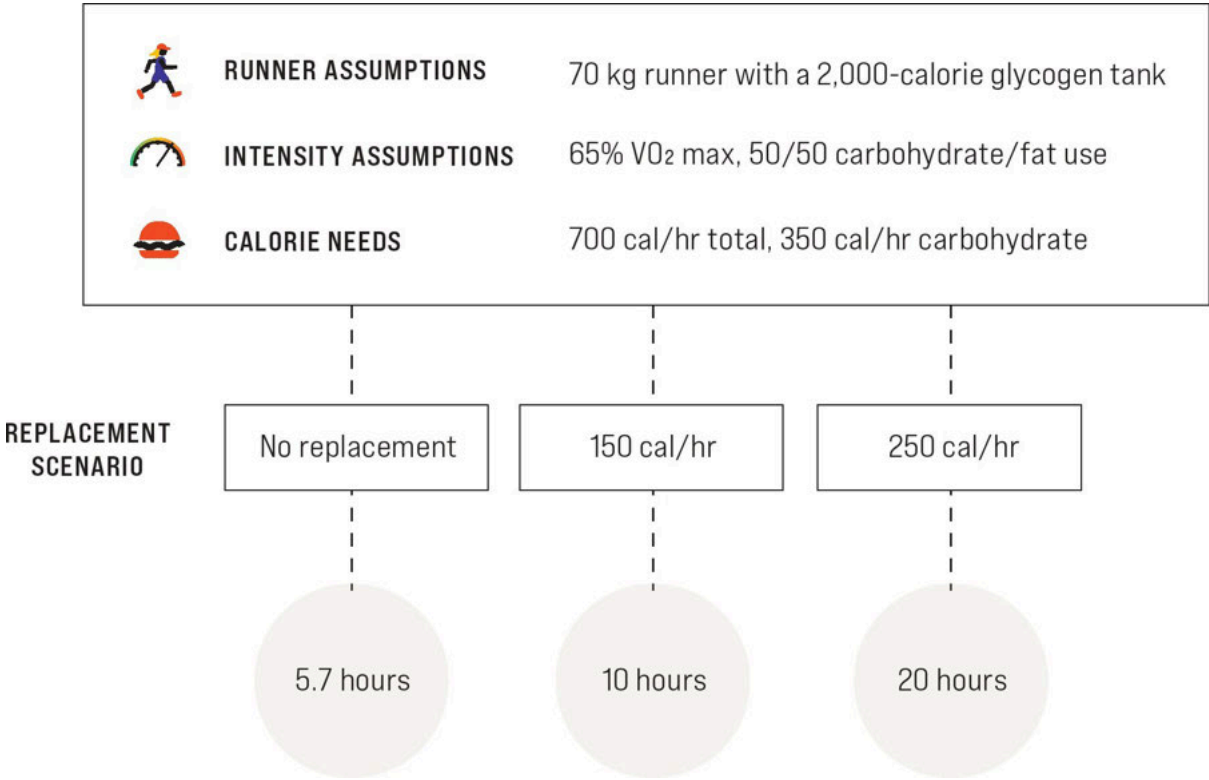


FIGURE 13.4 How long can you run with reasonable assumptions about intensity and caloric replacement?

So how much carbohydrate do you need for training and racing? The answer really depends on your performance goals. Generally speaking, the higher the intensity you need to maintain to accomplish your goal, the more carbohydrate you need. Subsequently, if you are a very fit runner interested in squeaking by the cutoffs, you will need less carbohydrate because you are operating at a lower percentage of your VO2 max. What I advise athletes to do is to start at 60 g of carbohydrate per hour and move up and down from there based on the intensity of the race and what they can tolerate.

JASON KOOP'S SECRET RICE BALLS

ONE OF THE GREATEST TRUTHS about sports nutrition is that even the best foods are useless if they stay in your pocket. You have to put those calories, electrolytes, and fluids into your body for them to do you any good. That means you have to like how they taste, how they smell, how they feel in your mouth, and how easy they are to unwrap and get down your throat. When I was a novice endurance athlete, I loved Krispy Kreme doughnuts, and one of my early experiments in sports nutrition was to cram three or four into a sandwich bag and squeeze them out a torn corner, as you would a carbohydrate gel. My tastes and my cooking skills have improved since then, and I developed two variations of a rice ball that meet the during-workout and during-competition nutrition guidelines in this chapter and have the taste, texture, and convenience characteristics that make them a go-to favorite for several of my athletes.

BACON AND EGG RICE BALLS

makes about 12 rice balls

1½ cups uncooked basmati rice

2 eggs

2 strips bacon

2 oz grated Parmesan cheese

Salt to taste

1. Cook the rice.
2. Scramble and cook the eggs.
3. Cook the bacon. Drain excess fat and chop.
4. Combine rice, eggs, bacon, cheese, and salt in a large mixing bowl.
5. Scoop small portions into sandwich bags and tie the ends off.

Per ball: Calories: 133 / Carbohydrate: 18 g / Protein: 4 g / Fat: 5 g / Sodium: ~160 mg

SWEET AND SALTY RICE BALLS (VEGETARIAN)

makes about 12 rice balls

1½ cups uncooked basmati rice

2 eggs

2 Tbsp honey

1–1.5 Tbsp soy sauce to taste

1. Cook the rice.
2. Scramble and cook the eggs.
3. Combine rice, eggs, honey, and soy sauce in a large mixing bowl.
4. Scoop small portions into sandwich bags and tie the ends off.

Per ball: Calories: 115 / Carbohydrate: 20 g / Protein: 2 g / Fat: 3 g / Sodium: ~125 mg

PROTEIN

Protein's role in ultramarathon events is interesting. Although its role in energy metabolism is limited, there may be a role for protein as a way to mitigate the effects of muscle damage incurred during a race, although the research findings on this are still equivocal (Knechtle et al. 2011). Particularly during training, athletes would be well served to ingest ~40 g of protein every four hours throughout the day to aid in muscle protein synthesis (i.e., rebuild the muscle that you broke down). When your training runs are over four hours in duration, this recommendation carries particular weight in order to make the most out of the runs and facilitate recovery.

FAT

Simply put, fat's purpose in ultramarathon racing and training is to maintain the satiety of foods. Ingestion of fat will not materially add to your body's ability to oxidize fat as a fuel. Additionally, most individuals cannot consume more than 30 g (270 calories worth) of commercially available fat supplements like MCT oil without gastrointestinal distress.

FLUID CONSUMPTION

In addition to making sure you're consuming enough calories to support your workouts, you also have to make sure your fluid intake is adequate. The easiest way to determine whether your fluid consumption is keeping up with your fluid losses is to weigh yourself without clothing before and after your workouts (excluding runs of greater than 3 hours, because water released as a byproduct of oxidizing fuel will confound the results). Any weight you lose during this time is due to fluid loss. Although there is some evidence that athletes can lose 2 percent or more of their body weight due to dehydration with no decline in performance, and in some cases with an increase in competitive performance due to reduced body weight, in training there is no benefit to exceeding 2 percent weight loss during workouts.

Over the years, the notion of a 2 percent loss of body weight negatively impacting performance has become almost ubiquitous. This notion helped drive the now outdated recommendation to "consume as much fluid as tolerable" and likely has contributed to the rash of hyponatremia cases in ultrarunning (more on that later). In more recent years, that notion has been challenged, and "drink to thirst" has become a common moniker, suggesting that your sensation of thirst is all you need to tell you when to drink. These polar opposite recommendations (drink all you can vs drink to thirst) have befuddled race directors and athletes alike. I can vividly remember being at the pre-race meetings of two different 100-mile races that were separated by only six weeks and having the medical directors of each give these opposing recommendations on fluid consumption.

The fact of the matter is that if you ever try either of these strategies in a race, you will easily see how fallible they are. Weighing yourself at an aid station will contain far more error than the changes you are looking for (<2 percent) due to the sweat, dirt, and gunk stuck to your clothing. And don't even get me started on how accurate a scale is when placed on a patch of dirt next to your drop bags. Similarly, anyone who has run the Badwater Ultramarathon can attest to the fact that your perception of thirst is totally messed up by the many hours breathing in dry, hot desert air combined with copious amounts of sand and dust kicked up from your roadside crew. Both of these in race ways of determining fluid replacement are ineffective, unsound, and likely to cause more harm than good in an ultramarathon setting. So, let's stop recommending them and get to a better solution.

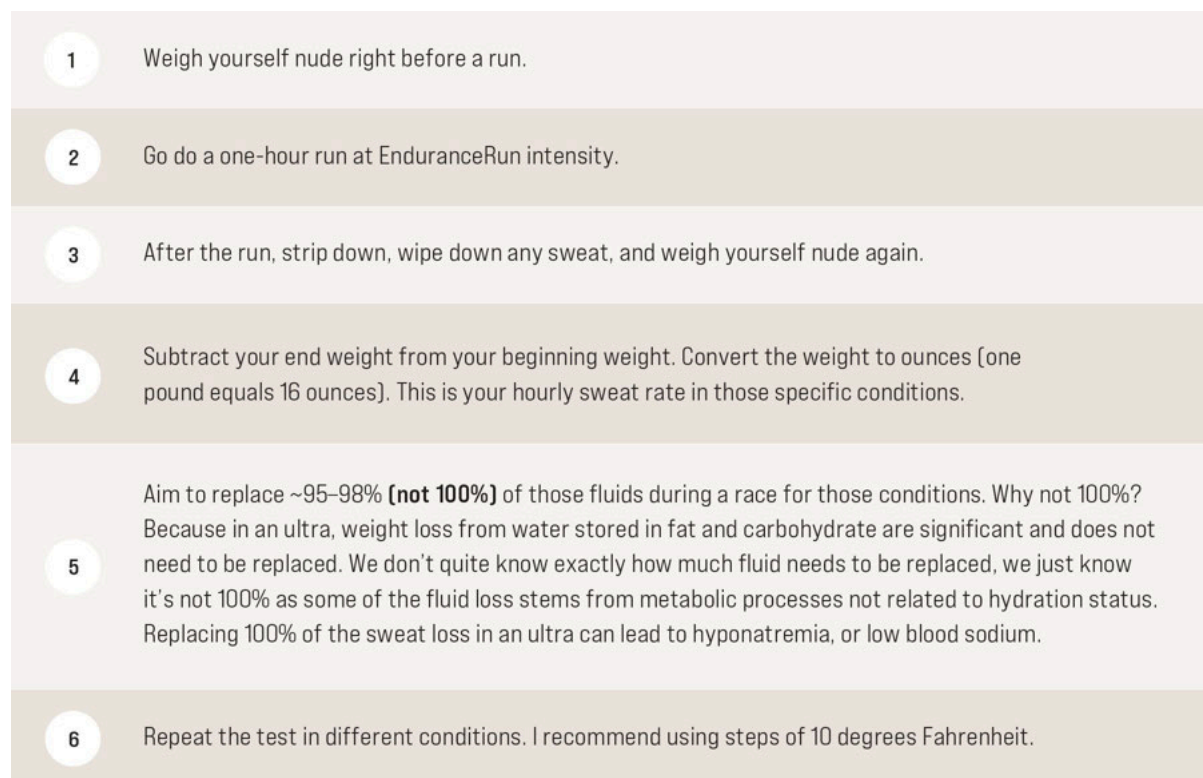


FIGURE 13.5 The Sweat Test.

The sweat test is a simple, effective tool you can use to get the best gauge on the fluids you will need to replace during training and racing. And the best thing about it is all you need is a scale.

Once you have done this across a range of conditions, it's easy to map your fluid replacement rate against the conditions of the race you are faced with by using a simple weather app that features hour-by-hour temperature conditions. It is worth noting that intensity affects your sweat rate as well (the more intense the exercise, the more you will sweat). However, given that most ultramarathons are performed around EnduranceRun intensity, the sweat test (Figure 13.5) covers a variety of ultramarathon scenarios and works as a good starting point.

SODIUM

Sodium is the last mission-critical nutrient in your racing and training arsenal. As with carbohydrate and fluid, sodium recommendations have ebbed, flowed, and changed over time. First off, the amount of sodium you require is a function of the amount of sodium lost in the sweat. Yes, you lose sodium through other sources, but sweat is the main one to focus on. Therefore, it's most useful to think about the amount of sodium you need to replace in terms of milligrams per unit of fluid you're ingesting to replace the fluid lost. The next step in the equation is determining the concentration of sodium in your sweat. As with any other biological process, there is a tremendous amount of individual variation for sweat sodium losses (see Figure 13.6). As a testament to this, we all know the "salty sweater" who is caked in dried sodium after a run. Aside from putting up with a little teasing and maybe an extra wash cycle for their clothes, salty sweaters will also need a bit more sodium in their ultrarunning routine. A more scientific version of the salt-caked shirt can be found with commercially available sweat sodium testing. If you think you are a candidate for this, I encourage you to work with a registered dietitian who works with endurance athletes.

Finally, it is important to understand that you do not need to replace 100 percent of the sodium lost in sweat, as you have sodium reserves located in the body, namely in the skin. Current sports science is a bit conflicted on how much sodium you need to replace, but our best estimates are anywhere from 50 to 80 percent of the sodium lost in sweat. So, even if you have a precise sweat sodium test performed, realize that the goal is not 1:1 replacement, but rather to just get it close.

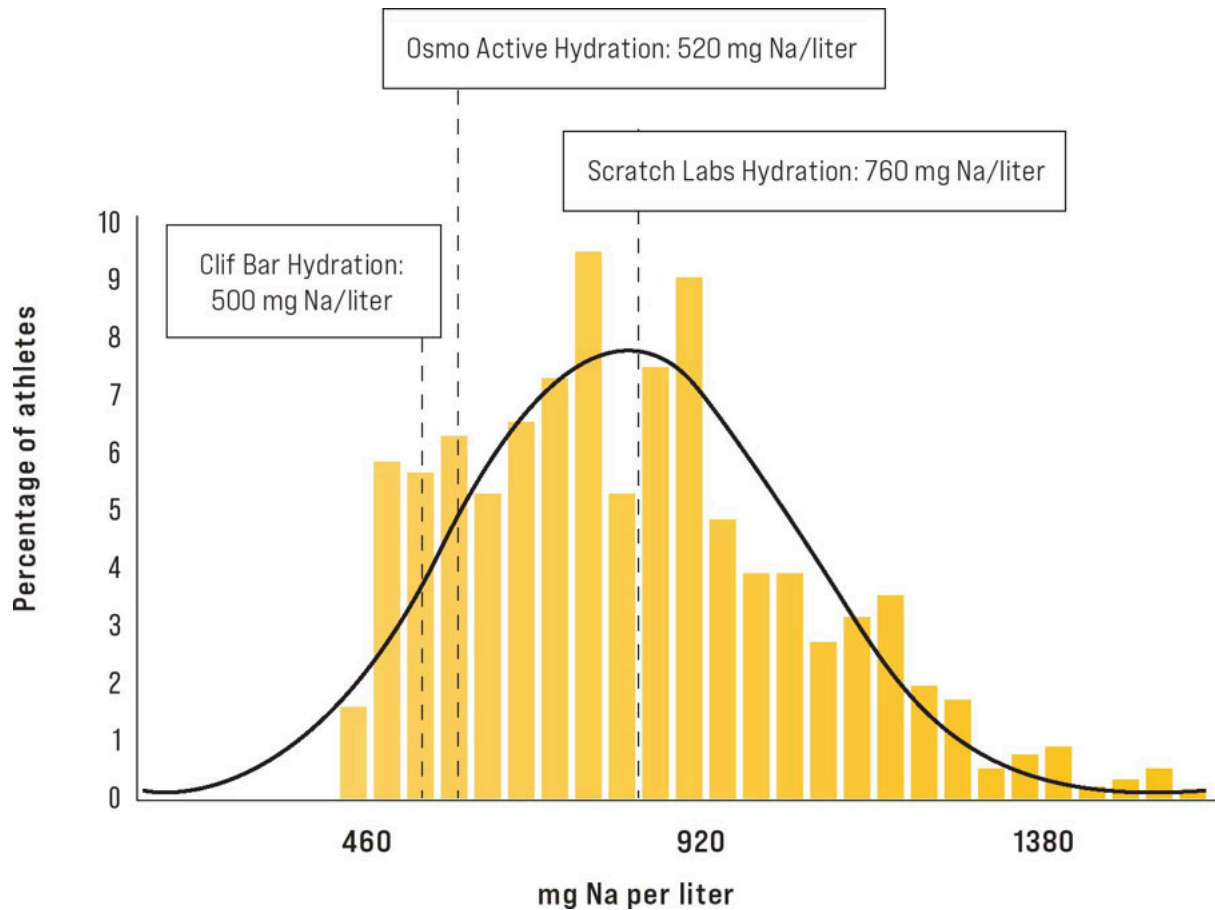


FIGURE 13.6 Typical sweat sodium concentrations. Adapted from Baker 2017.

Confused yet? Don't worry, sports drink manufacturers have made getting your sodium requirements easy for you, and most high-quality sports drinks will get you in the right ranges provided you mix the drink according to the manufacturer's recommendation. To meet your sodium needs, your goal should be to consume 600–800 mg of sodium per liter of fluid consumed. However, the sodium doesn't have to be in the drink; you can meet the requirement with plain water and sports nutrition products or real foods containing sodium, too. Even in the case where the sports drink you choose is a little light on sodium, it's pretty easy to get in the right range by adding normal sports nutrition products and other foodstuffs to your plan to make up the difference.

Lastly, what about those salt tablets? Salt tablets are generally not necessary and can encourage overdrinking due to the fact that they create an extremely hypotonic environment in your stomach (too much sodium, not enough fluid). Therefore, it's better to avoid them and get your sodium from a combination of electrolyte drinks, gels, chews, and real food.

WHAT THE HECK IS IN MY SPORTS DRINK?

A SPORTS DRINK IS ESSENTIALLY water with stuff dissolved in it. Some drinks have lots of different kinds of stuff dissolved in them, most of which just waste space. There is only so much room to dissolve solutes in a drink, and drinks with fewer ingredients can use more of that room for important things such as carbohydrate and sodium. The simplest drinks are the best because they are easiest on the gut and facilitate the transport of sugar and electrolyte across the semipermeable membrane of the intestinal wall better and faster.

The concentration of sports drinks is important. When you change the osmolality of the fluid (the total molecular concentration of everything in the drink—carbohydrate, electrolyte, flavoring, additives—per unit volume), it changes how the drink influences the overall mixture in your stomach and hence how that mixture makes it into the intestine. Sports drinks are formulated to optimize the absorption of carbohydrate, fluid, and electrolyte. If the osmolality of the sports drink is too high because of a bunch of additives, it may contribute to slower gastric emptying. When the osmolality of sports drinks is lower, it is more likely to contribute to faster gastric emptying (depending on what else you're eating and drinking), and if it's being consumed on an empty stomach it is formulated to get into the intestine quickly. If you are designing a sports drink to have a relatively low osmolality but you want it to deliver moderate to high amounts of sodium and/or carbohydrate, you have to eliminate anything that's not essential.

The primary ingredients, sugar, sodium, potassium, and flavoring, are in the drink for good reason. Putting electrolytes and flavoring into a fluid makes you want to drink more frequently and consume more fluid each time you drink. There's actually a lot more to the way your sports drink tastes than marketing mumbo jumbo. A lightly flavored drink is preferable to a stronger one because when you consume half a bottle in one long slug, the stronger-tasting drink becomes overwhelming and you stop drinking. A drink that tastes almost watered down when you are at rest will taste just about right when you are running. This is why athletes have long diluted commercial sports drinks like original Gatorade, which these days are often flavored to appeal to convenience store customers instead of athletes.

Even taste components and mouthfeel are important. A slightly tart drink will encourage you to drink more than an overly sweet one, and citrus flavors also increase the drive to drink. It should be no surprise, then, that almost every drink company has some version of lemon-lime and/or orange in its product line. In addition to the flavor, a sports drink needs to clear the mouth well. When a drink leaves a film in your mouth, as is often the case with overly sweet drinks, it's not only unpleasant, but you're not likely to drink again soon.

Rather than dilute sports drinks, it is better to find a drink with a lighter taste so you can comfortably consume it at full strength. If you are consuming a high-quality sports drink like Skratch Labs, Osmo, or Clif Hydration, it's generally unnecessary, and can be counterproductive, to dilute the drink.

Making Sure You’re Getting It Right

There’s a simple way you can make sure your carbohydrate, fluid, and electrolyte intake is on target during workouts. An exercise I use at ultrarunning camps is to have athletes hold on to the wrappers from everything they consume during a four-hour EnduranceRun. After the run, we lay them all out and record the following for each item: carbohydrate calories, milligrams of electrolytes, and milliliters of fluid. We then give the athletes a report card outlining the results, where they did well, and where they might have failed.

After tallying up everything you have consumed, do some simple math. You are aiming for 200–300 calories per hour (with ~90 percent coming from carbohydrates; you can also subtract the first hour of the run if you have just eaten a meal), 16 to 32 oz of fluid, and 600 to 800 mg Na per 32 oz (about a liter) of fluid. The purpose of this exercise is to uncover what you are actually taking in and where you are deficient. You then can adjust your ranges based on your personal calorie, fluid, and sodium needs. The fluid side of the equation will be of particular importance, since as mentioned earlier it will vary tremendously based primarily on the environment.

CALORIES IN YOUR POCKET, HYDRATION IN YOUR BOTTLES

WHAT SHOULD BE IN YOUR BOTTLES during workouts: water or a high-carbohydrate and electrolyte sports drink?

For the vast majority of runs, a high-carbohydrate drink is not necessary. While high-carbohydrate drinks have a place in sports nutrition, separating calories from hydration is a better and more versatile strategy. What’s in your bottle or hydration pack should serve the purpose of hydration, and the food in your pocket should serve the purpose of fueling.

As workload and temperature change, your fluid and calorie needs change independently. When it is hot, you need more fluid per hour, sometimes twice as much as during a cool run. This often happens during the course of an ultramarathon, where you start in the cool morning and run through the heat of the day. If your fluids contain 35 g or more of carbohydrate per 20 oz serving, you’ll consume 35 g if you drink one bottle and 70 g if you drink two bottles in one hour. That’s at the top end of the recommended range of 60 to 90 g of carbohydrate per hour; any further calories can lead to gastric distress. When you start to feel the initial stages of gastric distress, you delay drinking and start digging a dehydration hole that can take hours to recover from. Remember, you can come back from a caloric deficit easily and within minutes by eating. Returning to normal hydration status takes much longer, and being dehydrated during that time can have more deleterious effects on your performance.

Separating calories from hydration allows you to ratchet up fluid intake in response to high temperatures and dial it back in cooler conditions without greatly affecting your calorie supply. It also allows you to vary carbohydrate sources more easily because they are not tied to what you are drinking. This is important because your food choices can supply a large amount of the sodium you need to replenish. In many cases you can consume the recommended amount of sodium entirely through food sources while drinking plain water.

COSTA ET AL. 2019

CARBOHYDRATE RECOMMENDATIONS	Upwards of 90 g/hr (2:1 glucose to fructose) although such a high level likely unnecessary
FLUID RECOMMENDATIONS	Provided sufficient fluids/volumes are available, "drink to thirst," "ad libitum"
SODIUM RECOMMENDATIONS	Avoid excessive sodium supplementation during running. Consume sodium based on food cravings. Do not use highly visibl
OTHER RECOMMENDATIONS	Use trial and error with foods/fluid quality/quantity to determine what is optimal. Try tosimulate race day (and thus know v
	Use "B" or "C" races to determine how other contributing factors (travel, weather, pacing,competition, stress, changes in nc
	Consume slightly more early (first 2 hrs) as GI symptoms tend to develop later.
	In longer (greater than 8 hrs) races, avoid excess protein, fat, fiber, or FODMAP-heavyfoods; can mouth rinse with carbohy

CARBOHYDRATE RECOMMENDATIONS	30–50 g/hr
FLUID RECOMMENDATIONS	450–750 ml/hr (by drinking every 20 min), greater in hot and humid conditions
SODIUM RECOMMENDATIONS	> 575 mg/L
OTHER RECOMMENDATIONS	In training: Individualized, periodized, food-first approach. Moderate-to-high carbohydrate diet (~ 60% of energy intake, 5– Limit carbohydrate before occasional easy sessions and/or moderating daily carbohydrate intake, which may enhance fat ox In racing: 5–10 g/hr of protein. Eat (carbohydrate and protein) from variety of sources, more savory foods in longer races. U

ACSM (AMERICAN COLLEGE OF SPORTS MEDICINE) POSITION STATEMENT 2016

CARBOHYDRATERECOMMENDATIONS	Up to 90 g/hr while exercising (ultra specific), 6–10 g/kg/d (for endurance athletes, notultra specific)
FLUID RECOMMENDATIONS	Drink 5–10 ml/kg in the 2-4 hrs before exercise (pale yellow urine color). Drink enough during to limit day’s weight loss to
SODIUM RECOMMENDATIONS	Keep blood sodium above 135 mmol/L; doesn't say how to achieve it, nor is it ultra specific
OTHERRECOMMENDATIONS	Nitrates improve exercise tolerance, economy, and performance in at least non-eliteathletes.

TABLE 13.1 Nutrition recommendations for ultrarunning in scientific literature.

An example of this nutrition report card, detailed in Figure 13.7, is a from a runner returning from a four-hour run having consumed 64 oz of total fluid, including three bottles of drink mix, as well as two gels and two bars.

Preworkout Nutrition

The most important aspect of the last full meal you eat before a training session is that it's out of your stomach and digested before you start training. This is especially important for interval workouts because higher-intensity efforts tend to be downright unpleasant on a full stomach. A relatively light meal that's rich in carbohydrate (preferably about 70 percent of total calories) is a good choice because meals that contain a lot of fat or protein stay in the stomach longer and are digested more slowly. That's a good thing if you're trying to feel full longer, but it's not good if you're about to go out for a hard workout. Examples of good meal choices are pasta, a turkey sandwich, or oatmeal with fruit.

	INVENTORY	FLUID	CALORIES	SODIUM
HOURL 1	1 BoBo Bar (peanut butter)	16 oz water	330 cal	95 mg Na
HOURL 2	1 GU (vanilla bean)	16 oz Skratch Hydration	180 cal	440 mg Na
HOURL 3	1 Clif Bar (blueberry crisp)	16 oz Skratch Hydration	340 cal	560 mg Na
HOURL 4	1 Clif Shot (razz)	16 oz Skratch Hydration	180 cal	475 mg Na
TOTAL		64 oz	1030 cal	1570 mg Na
TARGET	16-32 oz / hour		200-300 cal/hour (MINUS FIRST HOUR)	600-800 mg/ 32oz fluid
GRADE	A B C D F		A B C D F	A B C D F

Overall Grade

A

FIGURE 13.7 A nutrition report card.

When it comes to your preworkout or pre-race meal, a good rule of thumb is that meal size should get smaller the closer to the workout or race you get. For instance, you can consume 1.5 g of carbohydrate per kilogram of body weight when your final preworkout meal is three hours before your run (Table 13.2), but this should be closer to 1 g/ kg if you're going to train two hours after your last significant meal. Also, you shouldn't try to fill your daily fiber requirement at this time. The American Heart Association recommends 25 to 30 g of fiber a day to reduce LDL cholesterol (the bad kind) and lower the risk of heart disease, but fiber slows digestion, so it's better saved for other meals.

Postworkout Nutrition

Postworkout nutrition might be the most oversold concept in all of sports nutrition. Early on in my coaching career, I relentlessly pushed recovery shakes and concoctions to my athletes after just about every workout. The theory was, the quicker your replaced what you lost, the more effective the workout and the more thorough your recovery. Furthering my bias was the fact that one of my coaching colleagues, Kathy Zawadzki, performed research that led to one of the original recovery drinks, Endurox R4. This drink was complete with what was thought at the time to be a magical ratio of 4 g of carbohydrate to 1 g of protein. The theory (again at the time) was that a beverage containing exactly 4 g of carbohydrate to 1 g of protein consumed within twenty minutes of exercise was the linchpin for attaining superhuman levels of recovery. (OK, maybe the fine folks over at Endurox didn't promise that you would leap over buildings in a single bound, but you get the picture.) While fluid, protein, sodium, and carbohydrate are important elements in your postworkout routine, we now know there's little need for such laser-guided precision.

BODY WEIGHT (KG)	3-4 HOURS PRIOR	2-3 HOURS PRIOR	1-2 HOURS PRIOR	0-60 MINUTES PRIOR
	1.5-2.0 g/kg	1.0-1.5 g/kg	0.5-1.0 g/kg	0.25-0.5 g/kg
55 (121 lb.)	83-110	55-83	28-55	14-28
60 (132 lb.)	90-120	60-90	30-60	15-30
65 (143 lb.)	98-130	65-98	33-65	16-33
70 (154 lb.)	105-140	70-105	35-70	18-35
75 (165 lb.)	113-150	75-113	38-75	19-38
80 (176 lb.)	120-160	80-120	40-80	20-40
85 (187 lb.)	128-170	85-128	43-85	21-43

TABLE 13.2 Carbohydrate recommendations prior to exercise.

In terms of carbohydrate, you want to start workouts and competitions with full glycogen stores. You can store about 1,600 to 2,000 calories of glycogen in your muscles and liver, with a bit more circulating in your blood. This carbohydrate energy will be burned alongside fat and a bit of protein in muscle cells as you exercise, but of the three, carbohydrate is the only one you can realistically deplete during a single bout of exercise, and it will certainly be depleted during an ultramarathon or a long training bout.

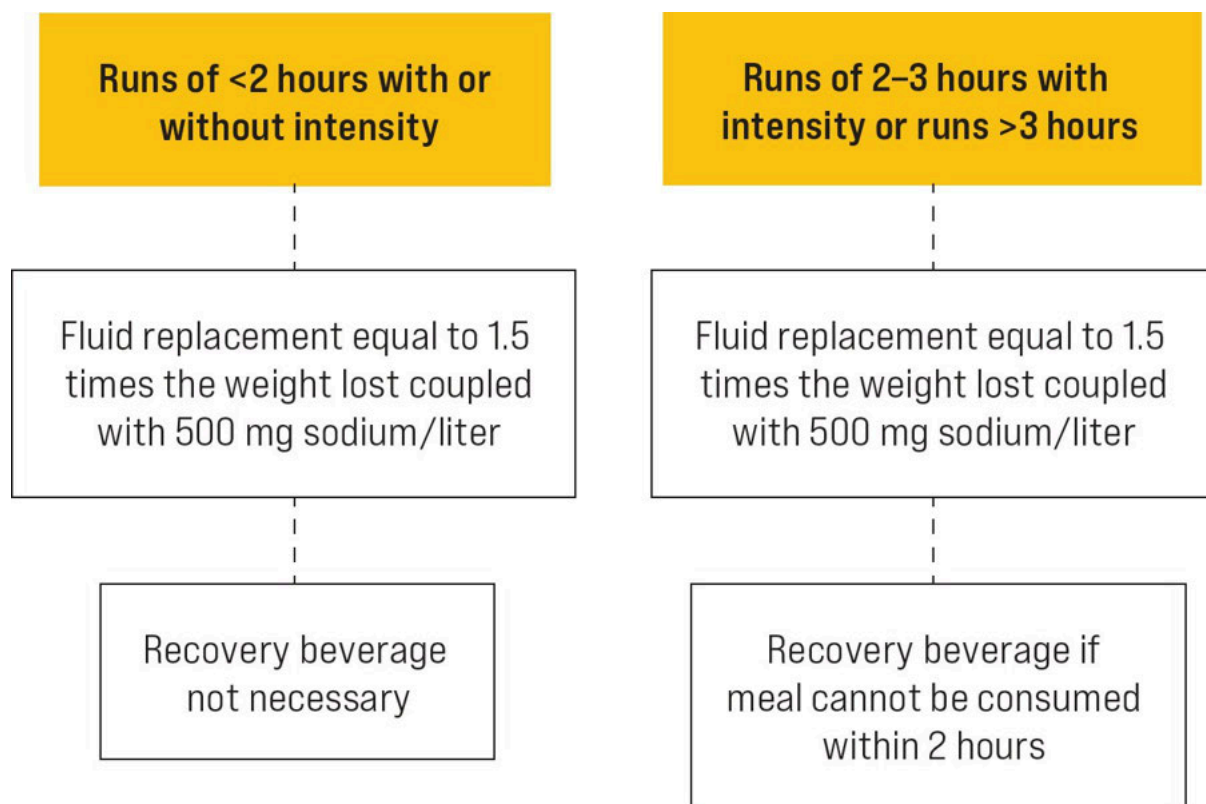


FIGURE 13.8 When a recovery drink is useful.

Fortunately it is relatively easy to ensure your glycogen stores are topped off before your next workout. Your body is primed to replenish muscle glycogen stores most rapidly within the first thirty to forty-five minutes following exercise. During this period, known as the “glycogen window,” there literally are more gates or “windows” open to allow sugar to enter muscle cells.

The glycogen window is important, but it is also important to keep replenishment in perspective. We used to think immediate postworkout consumption of carbohydrate was absolutely crucial for maximizing glycogen replenishment. Nutrition companies capitalized on this concept by designing specific recovery shakes and beverages that were touted to “optimize recovery,” and athletes started consuming these concoctions in large quantities, with little regard to their actual efficacy. While it is a good idea to get started with replenishment in the initial forty-five minutes after exercise because muscle cells are very receptive to taking in carbohydrate during that time, glycogen replenishment will reach 100 percent within twenty-four hours after exercise as long as you are consuming high-quality meals throughout the day. Therefore, I recommend that athletes optimize recovery with specific recovery beverages when both of the following conditions are met:

Your workout is over three hours OR your workout is over two hours and contains some structured intensity,

AND you don't have access to a high-quality meal immediately after the run (i.e., you have to drive for an hour to get home, or some other situation precludes you from sitting down to a good meal).

If you like drinking a postworkout shake just because you like it, that's great. Just don't think that it holds the key to your athletic future or that it is substantially more effective than eating a balanced meal.

Fluid and sodium are also important to replace post workout. Your goal within the first two hours after exercising should be to consume enough water to equal 1.5 times the water weight you lost during the session coupled with ~500 mg of sodium per liter of fluid. In other words, if you lost 2 lb (32 oz) during your workout, you should drink 48 oz of fluid within the two hours after you get back. This hydration rule of thumb applies to any run, not just runs over two hours or that contain intensity.

KEY POINTS FOR ESSENTIAL NUTRITION AND HYDRATION

You are constantly using carbohydrate, fat, and protein for fuel when exercising.

Protein is only used minimally for fuel, but it is important for repairing muscle damage, helping to maintain your immune system, and other necessary functions.

As you increase intensity, you will also increasingly rely on carbohydrate as a fuel.

Appropriate fluid is important for maintaining core temperature, gut motility, and blood volume.

You can check your day-to-day hydration status by using a combination of body weight, urine color, and thirst, all examined first thing in the morning.

Key nutrition targets for runs greater than two hours:

- **16 to 32 oz of fluid per hour, adjusted to your individual needs, environmental conditions, and intensity**

- **200 to 300 calories per hour, adjusted per your tolerance**

- **600 to 800 mg of sodium per 32 oz of fluid ingested**

You can test your nutrition plan by using the nutrition report card.

Recovery beverages are only beneficial in certain situations.

